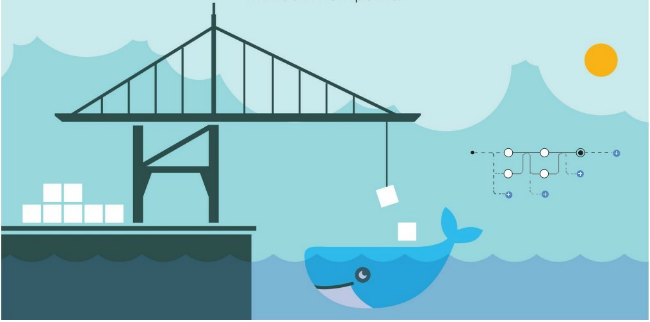
Integration of a Simple Docker Workflow with Jenkins Pipeline

This article is a tutorial on integrating the Docker workflow with Jenkins Pipeline.



In this article we will look at working with the pipeline script, Jenkinsfile, to which we will add the functionality to download a simple image from Docker Hub, build the image (identical copy), start a container off that image, run a simple test and, finally, if the test is passed, tag and publish the built image to the registry hosted in Docker Hub.

The prerequisites are:

- Jenkins v2.x standard install with the Pipeline plugin suite.
- A built agent setup capable of running Docker commands, configured as a node in the master.
- Basic knowledge of Groovy DSL to write pipeline code (either scripted or declarative syntax).

Plugins

Other than the set of plugins bundled as part of a Jenkins installation, ensure the plugins listed below are available as well, as they are essential to run the pipeline script that we will develop in this article.

- *https://plugins.jenkins.io/docker-commons*
- *https://plugins.jenkins.io/docker-workflow*
- *https://plugins.jenkins.io/docker-traceability*
- *https://plugins.jenkins.io/credentials-binding*

All the code snippets shown in the example below follow the declarative syntax as it's easier to get started with pipeline-as-code, which is especially attractive to beginners. Following this syntax, the complete pipeline-as-code is contained within a template that follows this pattern.

```
pipeline {
    agent, environment, options, parameters go here
    various stages to execute build go here
build commands or script to perform various
tasks go here
      tasks related to post-build go here
}
```

It's good to explore the links given below for a quick glance at pipeline syntax, as well as for the steps and examples. They are quite handy and serve as a great reference.

- *https://jenkins.io/doc/pipeline/steps*
- *https://jenkins.io/doc/book/pipeline/syntax/*
- *https://github.com/jenkinsci/pipeline-examples*

Docker support

Let us begin by looking at the agent section which has the provision, amongst others, to support the Docker build in